Final Project for SW Engineering

Class CSC648/848 Fall 2023

Team 02

California SafetyNet

Lukas – lheber@sfsu.edu – Team Lead, Github Master

Marshall – back end lead

Ely, Mark, Sanjith

Demo: California SafetyNet

Date: 12/5/2023

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1. Product summary

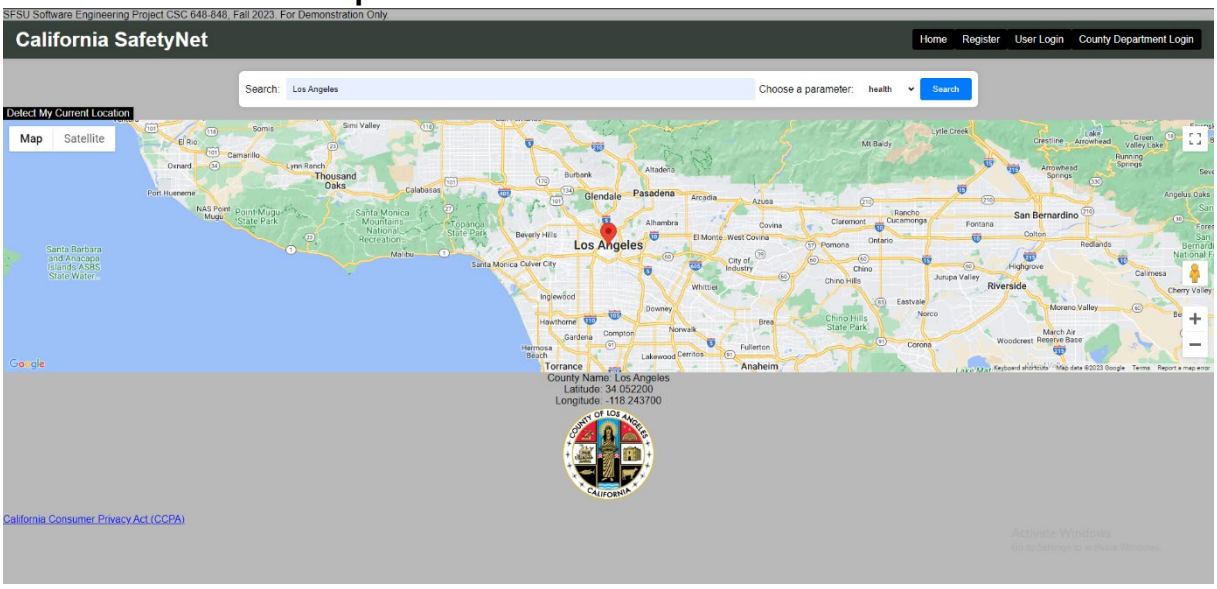
California Safety Net

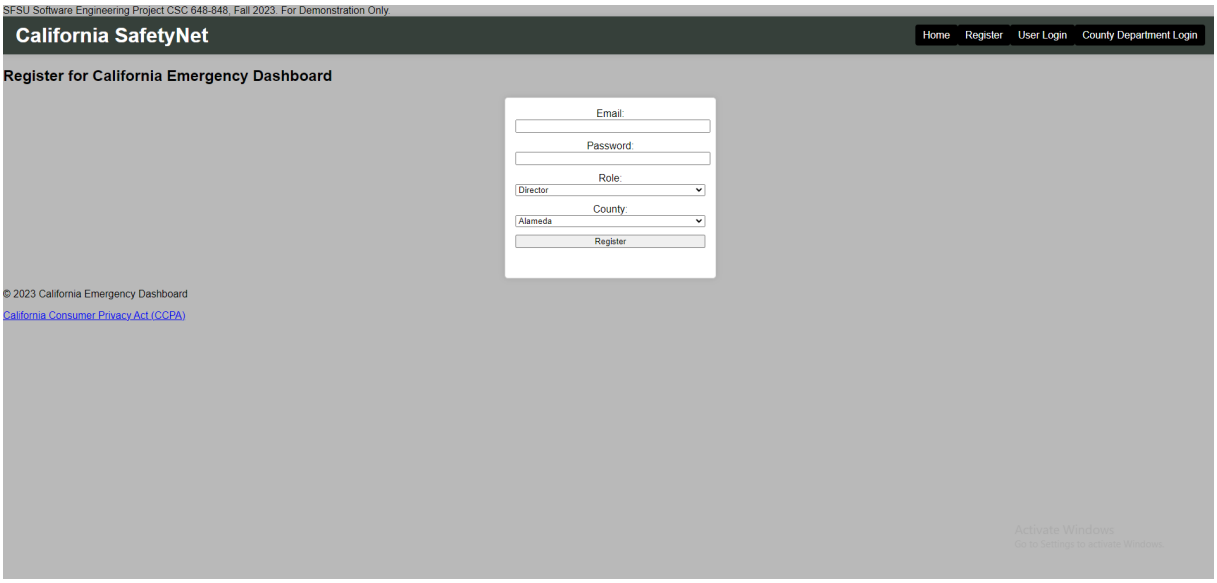
- Potential users shall be able to sign-up to the site and if needed shall then be authorized by administrators as official figures that can initiate alerts
- Registered and authorized users shall be able to initiate alerts to the general public in one “their” category, which is one of the following four: 1. Security, 2. Health, 3. Weather and 4. Fire
- Users shall be able to access a map that allows an overview of alerts for different counties
- Besides the map users shall be able to search and filter alerts to get an overview for example for planning a trip inside the state of California

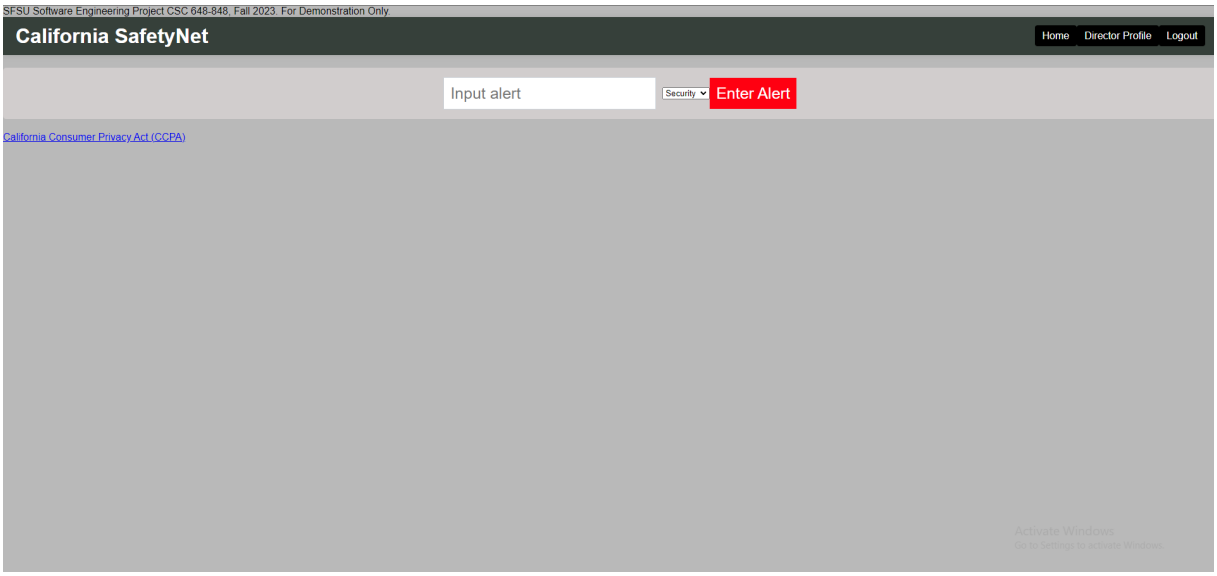
California Safety Net allows its users an easy way to stay safe and informed. Registered users are alerted by county officials should it be necessary. This helps to always keep the Californian public safe. A big aid in this is the extensive map feature that allows users to get an overview of the whole state of California and its safety situation. This enables planning trips during the fire season and provides a feeling of safety to its users. Fall into the reassurance provided by California Safety Net today!

Link to California Safety Net: California SafetyNet (softwareengineering.servehttp.com)

2. Screenshots of final product







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California SafetyNet

Home | User Profile | Logout

County Name: Alameda
Latitude: 37.648100
Longitude: -121.913300

Alert Type: General
Alert Description: Security alert for CountyA
Alert Date: 2023-10-10T18:36:54.000Z

Alert Type: Weather
Alert Description: Weather alert for CountyA
Alert Date: 2023-10-10T18:36:54.000Z

User Information

California Consumer Privacy Act (CCPA)

Activate Windows
Go to Settings to activate Windows.

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California SafetyNet

Home | Register | User Login | County Department Login

User Login

Email Address: Password: Login

California Consumer Privacy Act (CCPA)

Activate Windows
Go to Settings to activate Windows.

3. Key database tables

County_ID	County_Name	Latitude	Longitude	Image_Path
8	Calaveras	38.196000	-120.680500	./CalaverasCountyEmblem.png
9	Colusa	39.178200	-122.233700	./ColusaCountyEmblem.png
10	Contra Costa	37.919700	-121.930500	./ContraCostaCountyEmblem.png
11	Del Norte	41.743100	-123.897900	./DelNorteCountyEmblem.png
12	El Dorado	38.778400	-120.527000	./ElDoradoCountyEmblem.png
13	Fresno	36.985900	-119.232100	./FresnoCountyEmblem.png
14	Glenn	39.598900	-122.393200	./GlennCountyEmblem.png
15	Humboldt	40.745000	-123.869500	./HumboldtCountyEmblem.png
16	Imperial	32.977800	-115.559000	./ImperialCountyEmblem.png
17	Inyo	36.511300	-117.406000	./InyoCountyEmblem.png
18	Kern	35.343800	-118.727900	./KernCountyEmblem.png
19	Kings	36.075100	-119.815500	./KingsCountyEmblem.png
20	Lake	39.100600	-122.823500	./LakeCountyEmblem.png
21	Lassen	40.551200	-120.712000	./LassenCountyEmblem.png
22	Los Angeles	34.052200	-118.243700	./LosAngelesCountyEmblem.png
23	Madera	37.251900	-119.696300	./MaderaCountyEmblem.png
24	Marin	38.083400	-122.763300	./MarinCountyEmblem.png
25	Mariposa	37.579700	-119.907000	./MariposaCountyEmblem.png

Table 1: County

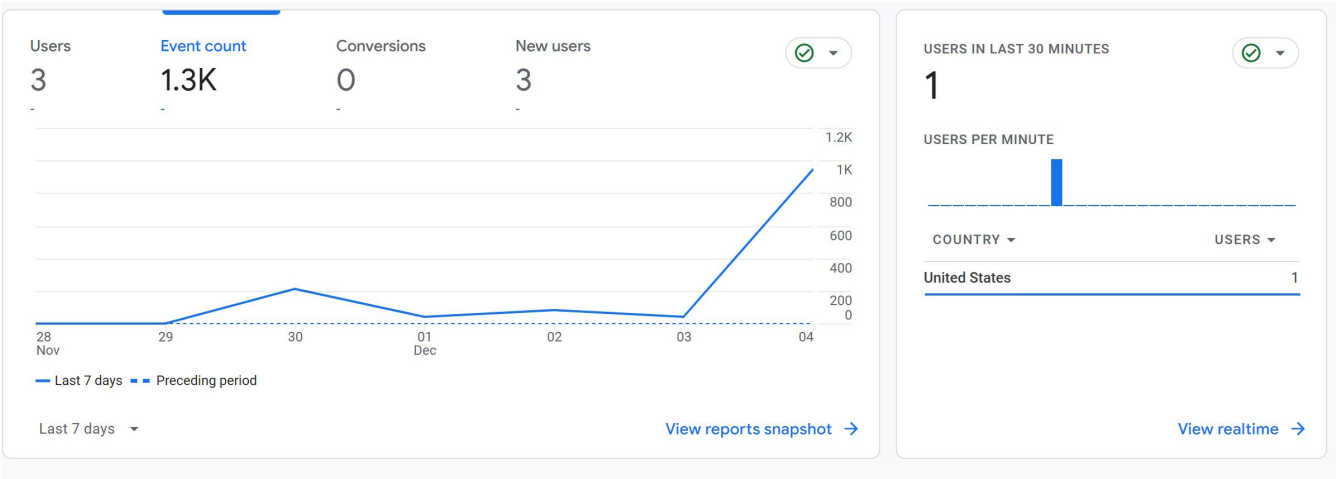
User_ID	Password	Email_Address	Phone_Number	Role	County_ID	Verified
UaArwCz7caZLiPjL4W6Bl8ej0Ae2	UkVEQUNURUQ=	mfeuer@mail.sfsu.edu	1234	P1	4	1
tp0a2K9X0SVoMqZkDZyEAPVvyBp1	UkVEQUNURUQ=	lukas27@mail.com	1234	P1	4	1

Table 2: User

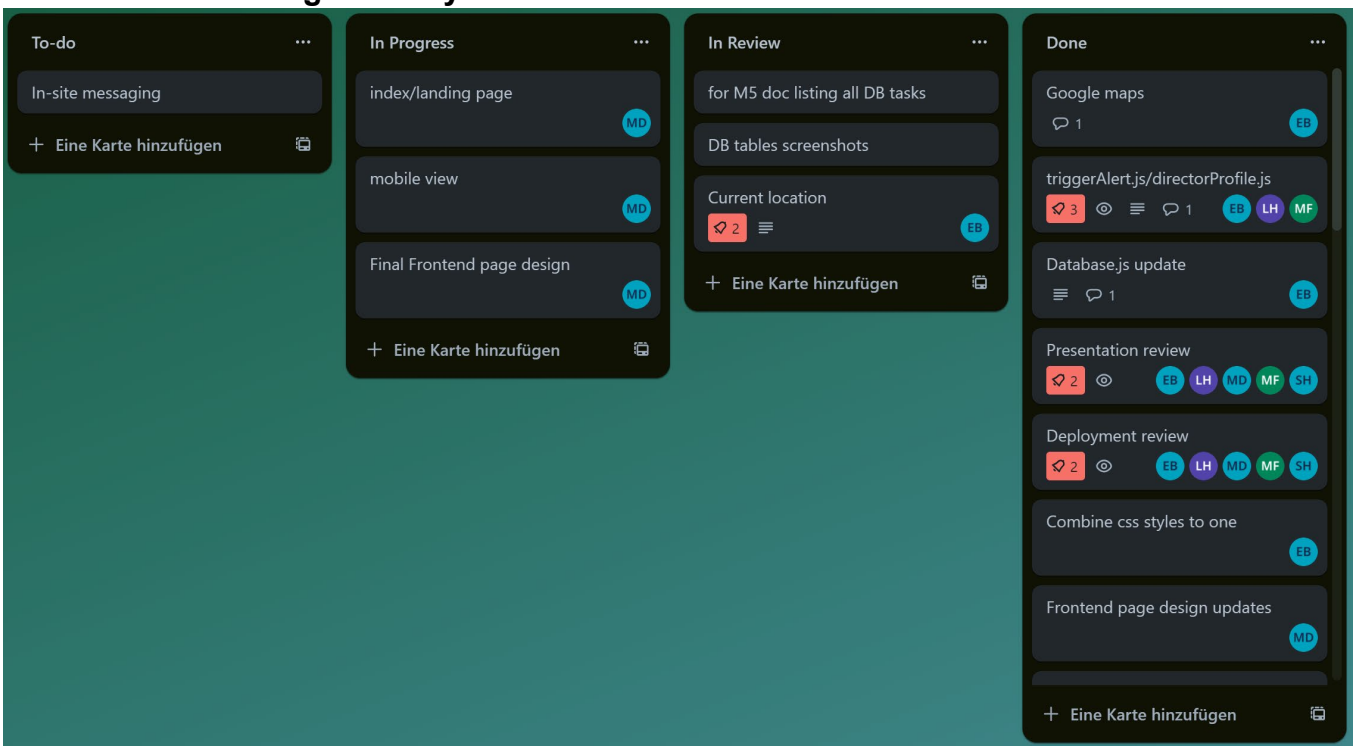
Alert_ID	Metric_ID	Alert_Message	Triggered_By	Timestamp	County_ID
1	1	Security alert for CountyA	1	2023-10-10 18:36:54	4
2	2	Weather alert for CountyA	2	2023-10-10 18:36:54	4
3	3	Health alert for CountyB	3	2023-10-10 18:36:54	5

Table 3: Alert

4. Google analytics plot



5. Use of task management system



6. Team member contributions

LH

Lukas Heber

An: Sanjith Raghavendra Hadagali; Ely Lee Barrientos; **+2 weitere** Mi, 29.11.2023 20:57

Hi there!

This is my list of contributions for our project this semester:
- Setup and maintained the AWS instance including the web server, the database and everything else
- Scheduled and lead all meetings
- Took over the github master role after the initial setup, created policies and branches to support a modern software development process
- Maintained the teams trello board including motivating the team members to uitilize it
- Helped team members with coding including extensive help via calls and dictating code to them, but also teaching them techniques for creating better (or working) code (SSH principles, maintainable code, security)
- Created M5, M4 and M3-feedback documents entirely on my own, organized the creation of all other Milestone documents including creating the actual final version and implementing the feedback given

If you disagree on anything I listed please let me know!

Best,
Lukas

Sanjith Raghavendra Hadagali

SH

Sanjith Raghavendra Hadagali

An: Lukas Heber; Marshall Theodore Feuer; Ely Lee Barrientos; Sa, 02.12.2023 05:34
fcs4manila@gmail.com

Contribution

- Design of all the Database tables.
- The Population of Database with Data
- Design of all the relevant database queries
- Testing of all the Queries on both the Local and EC2 database.
- Modifications to Database design as and when required.
- Incorporating more select queries based on team input.
- Testing example node Js program for database queries.
- Handling of png file in database table.
- Helping with UI tasks.
- Adding California privacy link and disclaimer.

Thank you! | I am interested. | Got it, thanks!

Antworten | Allen antworten | Weiterleiten

MF

Marshall Theodore Feuer

An: fcs4manila@gmail.com; Lukas Heber; Ely Lee Barrientos; Sanjith Raghavendra Hadagali

Do, 30.11.2023 21:56

- Wrote the triggerAlert.js function.
- Wrote/edited the JS code within script tag on index.html.
- Wrote/edited the search.js function.
- Wrote/edited the database.js function.
- Wrote/edited the server.js function.
- Organized 1 on 1 meetings with other team members to ensure productive workflow.
- Wrote code to add search parameter dropdown functionality.

Thank you! | Got it, thanks! | Thank you very much!

Antworten | Allen antworten | Weiterleiten

30.11.2023 - 14:04

Contributions

Von Mark D

Vollansicht

An lukas.heber@gmx.de | mfeuer@mail.sfsu.edu | ebarrientos1@sfsu.edu | shadagali@sfsu.edu

-Created html files
-Worked on the CSS
-Worked on the front-end design layout of each page
-Worked/Helped with the password salt and hash
-Worked on mobile view (in progress)

EB

Ely Lee Barrientos

An: Lukas Heber

Fr, 01.12.2023 23:37

Cc: fcs4manila@gmail.com; lukas.heber@gmx.de; Marshall Theodore Feuer; Sanjith Raghavendra Hadagali; Ely Lee Barrientos

-M0 helped build initial GitHub repository later taken over by Lukas
-M1 list of functional/non-functional requirements
-M2 high level UML Diagrams/Architecture and Organization
-M3
    -worked with Team to update search to populate all results and specified search
    -worked with Team to resize county image to 150x150pixels
    -worked with Team to update and realign server.js, database.js, search.js (ie. prevent database from duplicating tables, connection to server, connection database)
    -help rebase backend and frontend commits
-M4 Lukas covered, I missed this milestone due to work commitments
-M5
    -implemented google maps feature
    -worked with Team to update directorProfiles.html/database.js/triggerAlert.js/alert function to input into database
    -helped combine CSS style sheet
    -helped to update page layouts after combining CSS style sheet
    -pending work with Team to implement login/registration
-Attended Team meetings and 1:1 meetings to problem solve blockers
-Reviewed and approved pull requests
-11commits

7. Post analysis

Collaborating with the team on the project posed a challenge given the diverse skill levels within our group. However, varying and sometimes lacking levels of proficiency in the chosen technologies became a recurring hurdle for the team throughout the project.

In an attempt to empower individuals within the team, I experimented with providing more autonomy for independent work and pacing. Regrettably, this approach resulted in a notable slowdown in our progress, with competing priorities from other classes taking precedence for some team members.

Another big issue was scheduling meetings that everyone could attend due to work, other classes and other responsibilities. This wasn't helped by team members regularly missing our normal class time and/or our other meetings sometimes without any note. I don't really see a way to counter that behavior with the power that was given to me by the mode of the class, so I don't have an idea how I would counter this behavior in the future.

Contrary to these challenges, engaging in team discussions and conducting brief training sessions emerged as very valuable experiences, shedding light on the unique dynamics within our group. Moreover, transitioning to a scrum schedule, particularly the daily scrum meetings, played a pivotal role in enhancing accountability for the progress being made and fostering transparency and knowledge sharing among team members. This positive outcome further solidified my appreciation for agile methodologies, particularly scrum and was definitely the right thing in our situation.

Additionally, taking on too many tasks, particularly server maintenance, proved counterproductive, as it left me as the sole team member proficient in this area. In future situations, I would opt for collaboration with at least one other team member to distribute knowledge and responsibilities more effectively.

Finally, I should have taken over more of the coding responsibilities after discovering the issues my team members had, this would have allowed us to progress faster as a team and lead to a better result. This is definitely a big error of judgment on my part since I didn't realize that that would be necessary in our situation.